Graph Neural Networks with Imbalanced Data Distribution

[Appendix]

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The source code of this research paper has been made publicly available at **https://github.com/zwt233/BIM**.

1 SOCIAL INFLUENCE MAXIMIZATION

The goal of social influence maximization is to select a subset of $\mathcal{B}$ influential seed nodes that can maximize the influence propagation in a social network [1]. Specifically, given the full node set $\mathcal{V}$ and the seeding budget $\mathcal{B}$, the node selection process can be formulated as:

$$\max_{S} \ |\sigma(S)|, \ \textbf{s.t.} \ S \subseteq \mathcal{V}, \ |S| = \mathcal{B}, \tag{1}$$

where $\sigma(S)$ is a set of nodes influenced by the seed set S under the influence propagation models, such as linear threshold (LT) and independent cascade (IC) models [4]. This node selection process in Eq. 1 is known to be NP-hard, and a greedy algorithm can provide an approximation guarantee of $(1 - \frac{1}{e})$ if $\sigma(S)$ is nondecreasing and submodular with respect to S [5].

Recently, a line of work proposes to maximize the influence while balancing the influence for the different users from subpopulations. For fairness in influence maximization, [6] modifies the classic influence maximization problem with additional fairness provisions based on legal and game theoretic concepts. Besides, [2] studies the trade-offs between enforcing fairness and the loss of total influence, and [3] introduces the balance constraint as a two-stage problem. Although we also propose to maximize the influence, our work differs from these previous studies in two perspectives. 1) Different motivations of influence balance. Previous works are proposed to address fairness-aware influence maximization problems, while our method is specifically to balance the influence from a different class of labeled nodes. 2) Different utilizations of influence balance. As explained in Section 3.2, we especially defined a new influence balance concept for GNN in the semi-supervised node classification setting.

Following these previous work, we propose to maximize the number of influenced nodes and make these influenced nodes balanced in our BIM framework.

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